

The following heuristic algorithms are **not** guaranteed to find an optimal solution to the bin-packing problem, but they are fairly easily programmed on a computer.

- **Next fit algorithm:** we take the objects in the order they are listed, and place it in the current bin, if there is room. If there is not room, we start a new bin
- **First fit algorithm:** we take the objects in the order they are listed, and put each in the leftmost bin that has room. This is very easy to do, but does not give a very good result.
- **First fit decreasing algorithm:** we pre-process the list, putting the objects in decreasing order of size, and then use the first fit algorithm. This often gives a better result.
- **Full bin algorithm:** we search for combinations that produce full bins, and use the first fit algorithm on the remaining objects. This is the most time-consuming method, and not practical for large numbers of objects, but generally gives a very good result.

Question 1

32 45 17 23 38 28 16 9 12 10

The numbers in the list above represent the lengths, in metres, of ten lengths of fabric. They are to be cut from rolls of fabric of length 60 m.

- (a) Calculate a lower bound for the number of rolls needed.
- (b) Use the first-fit bin packing algorithm to determine how these ten lengths can be cut from rolls of length 60 m.
- (c) Use full bins to find an optimal solution that uses the minimum possible number of rolls.

Question 2

29 52 73 87 74 47 38 61 41

The numbers in the list represent the lengths in minutes of nine radio programmes. They are to be recorded onto tapes which each store up to 100 minutes of programmes.

- (a) Obtain a lower bound for the number of tapes needed to store the nine programmes.
- (b) Use the first-fit bin packing algorithm to fit the programmes onto the tapes.
- (c) Use the first-fit decreasing bin packing algorithm to fit the programmes onto the tapes.

Question 3

A small car-ferry has four lanes, each 20 m long. The following vehicles are waiting to be loaded. Can all these vehicles be taken on one trip?

Petrol tanker	14 m	Car and trailer	8 m
Car	4 m	Car	3 m
Range Rover	5 m	Coach	12 m
Car	4 m	Lorry	11 m
Car	3 m	Car	4 m
Van	4 m	Lorry	10 m

Question 4

Let the weights of the items be $a_1, a_2, \dots, a_k$ and the capacity of each bin be V . Explain why $\left\lceil \frac{\sum a_i}{V} \right\rceil$ gives a lower bound on the number of bins required. Give an example to show that it may not be possible to achieve this lower bound.

Question 5

Find a list of weights where first-fit gives a *better* packing than first-fit-decreasing.

Question 6

Prove that the first-fit algorithm uses no more than twice the optimal number of bins. Find a case where first-fit gives a solution very far from the optimal one.

Question 7

Surprisingly, it is possible to find a list of weights where *removing* an item leads to the first-fit-decreasing algorithm calling for a *larger* number of bins. Can you find such a list?

- **Best fit algorithm:** we take the objects in the order they are listed, and put each in the fullest bin which will accommodate it.
- **Worst fit algorithm:** we take the objects in the order they are listed, and put each in the emptiest bin which will accommodate it.
- **Almost-worst fit algorithm:** this is a variant of worst fit – if an object fits in more than one bin, we place it in the second-emptiest bin that will accommodate it

Question 8

Compare the performance of these heuristic algorithms when applied to the items $\{4, 8, 5, 1, 7, 6, 1, 4, 2, 2\}$ to be packed into bins of size 10.